Module 4: Multiple Linear Regression

Other Considerations Part 2

Wenting Xu

Iowa State University

Other considerations/potential issues in MLR

- Dealing with categorical or qualitative predictors.
- Multicollinearity.
- Non-constant variance of error.
- Non-linear relationships.

Assumptions for linear regression

- Relationship between Y and $X = \left(X_1, X_2, \ldots, X_p\right)$ is approximately linear.
- $E(\epsilon) = 0$.
- $\mathbf{Var}(\boldsymbol{\epsilon}) = \boldsymbol{\sigma}^2$. **Homoscedasticity**
- ϵ 's are uncorrelated.

Non-constant variance of error terms

- The standard errors of our estimates rely on this assumption.
- Additionally, carrying out hypothesis tests, constructing prediction intervals, and confidence intervals associated with the linear model also rely upon this assumption.

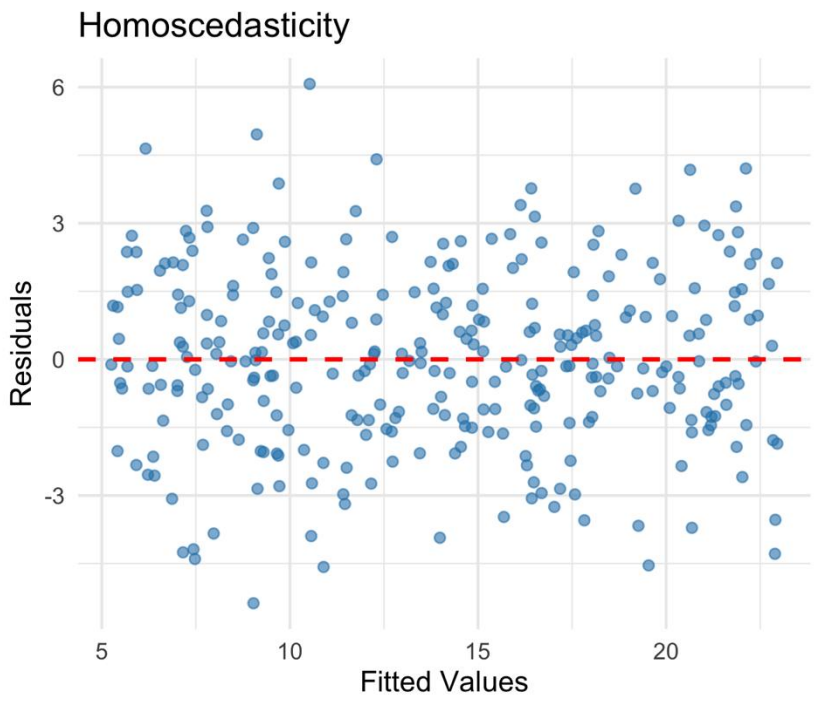
Non-constant variance of error terms

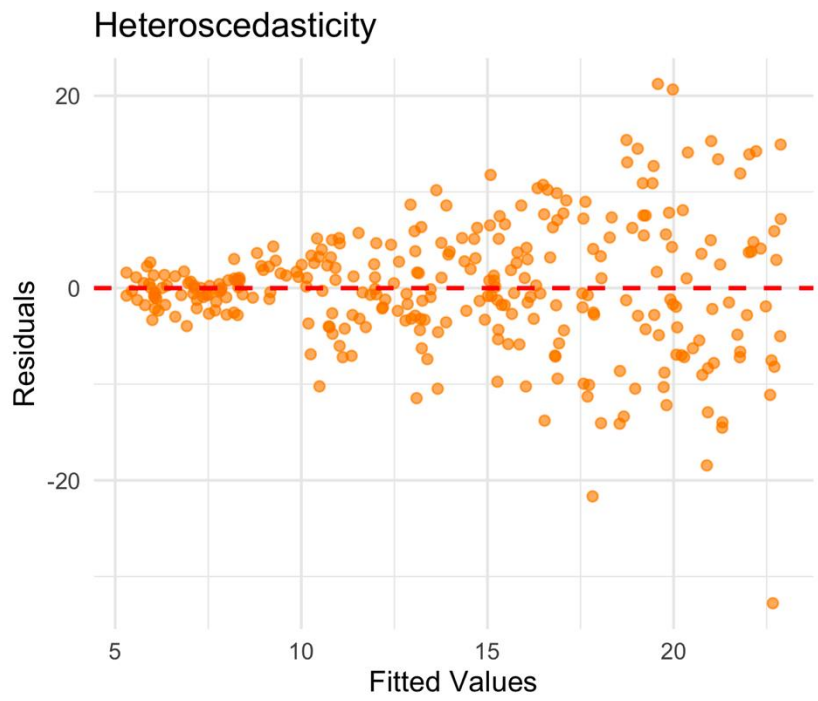
- It may be the case that the variances of the error terms are **non-constant**.
- For example, the variances of the error terms may increase with the value of the response.
- How might we identify whether this is a problem with our model?

Residual plot

To diagnose this, we can plot **residuals** (e_i) **vs. fitted values** $(\hat{y}_i)$ from our model.

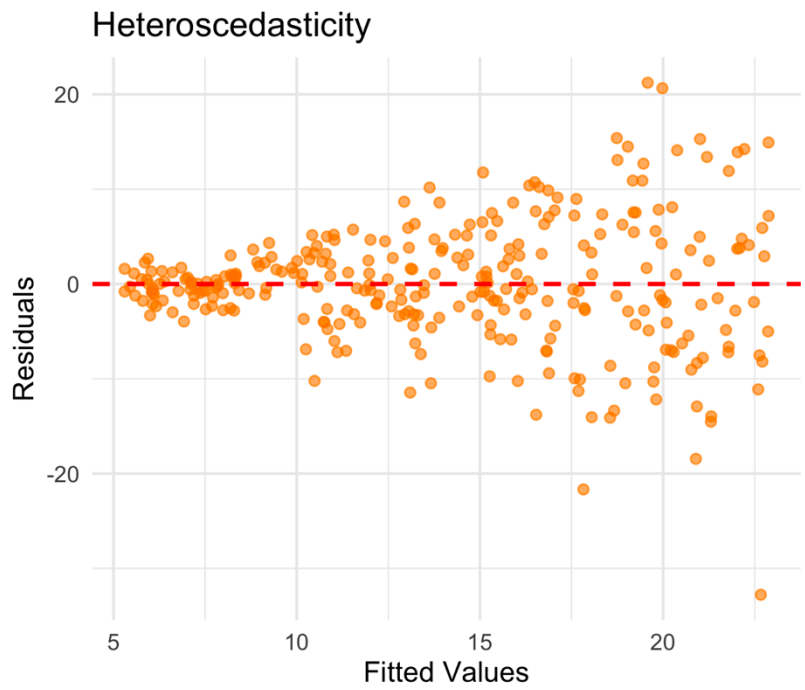
- **If the constant variance assumption holds**, the plot should exhibit random scatter (no discernible pattern).
- If the plot **funnel shape**, there's a problem.

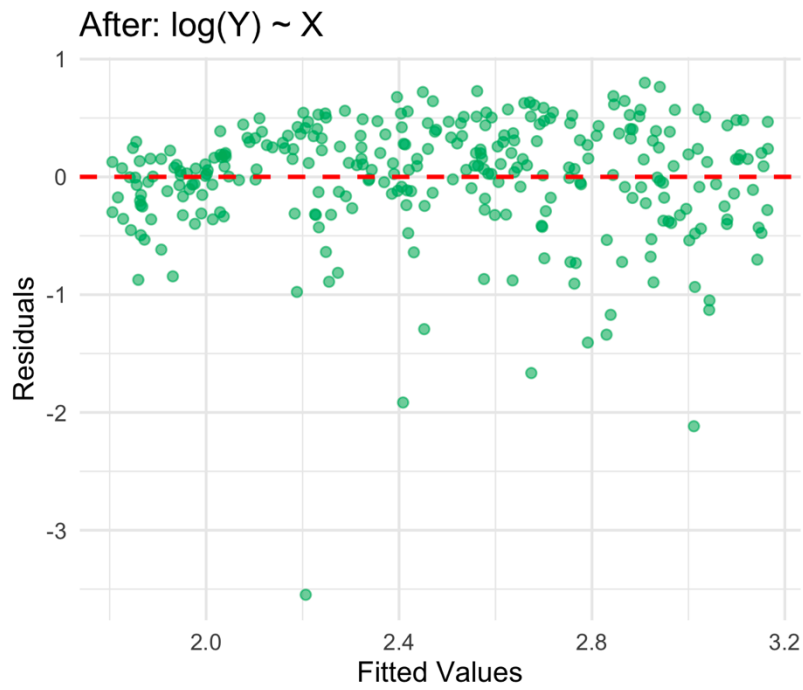




Non-constant variance of error terms

One possible solution: transform the response Y using a concave function such as $\log Y$ or $\sqrt{Y}$.





Other considerations/potential issues in MLR

- Dealing with categorical or qualitative predictors.
- Multicollinearity.
- Non-constant variance of error.
- Non-linear relationships.

How to diagnose non-linearity

- The linear regression model assumes that there is **a straight-line relationship between the predictors and the response.**
- If the true relationship is far from linear, then virtually all of the conclusions that we draw from the model are suspect.
- Additionally, the prediction accuracy of the model can be significantly reduced.

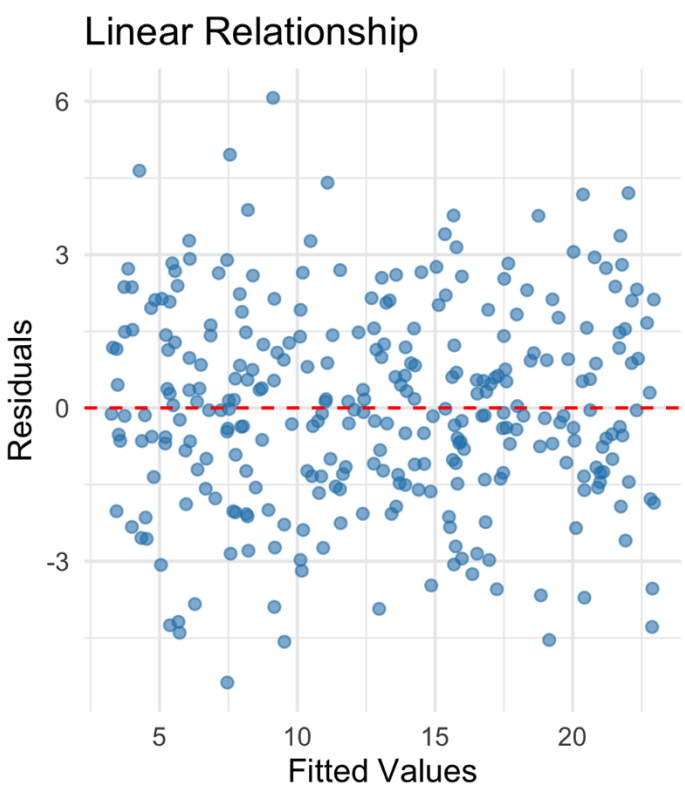
How to diagnose non-linearity

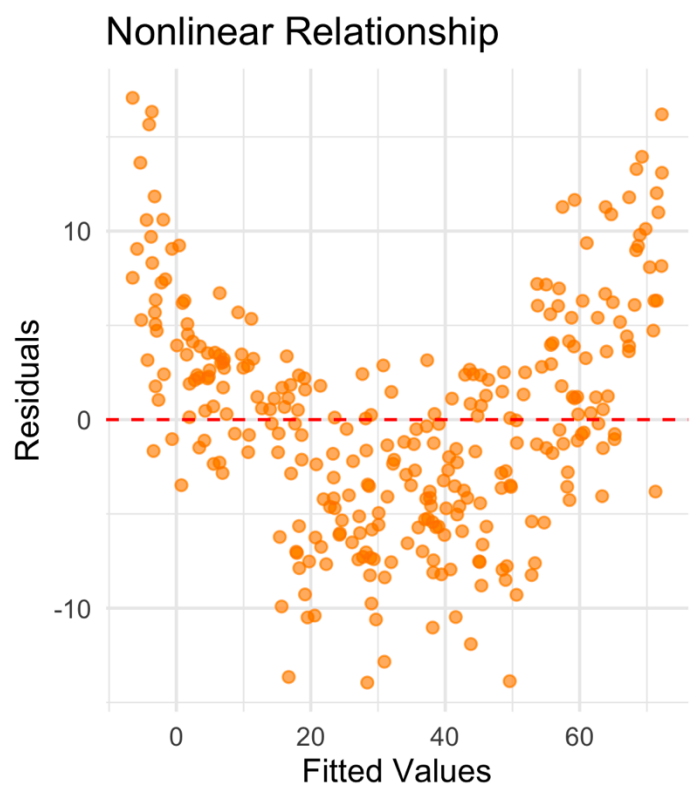
Suppose we have $p = 10$ predictors.

How might I diagnose non-linearity in this setting?

Residual plots: e_i verus $\hat{y}_i$

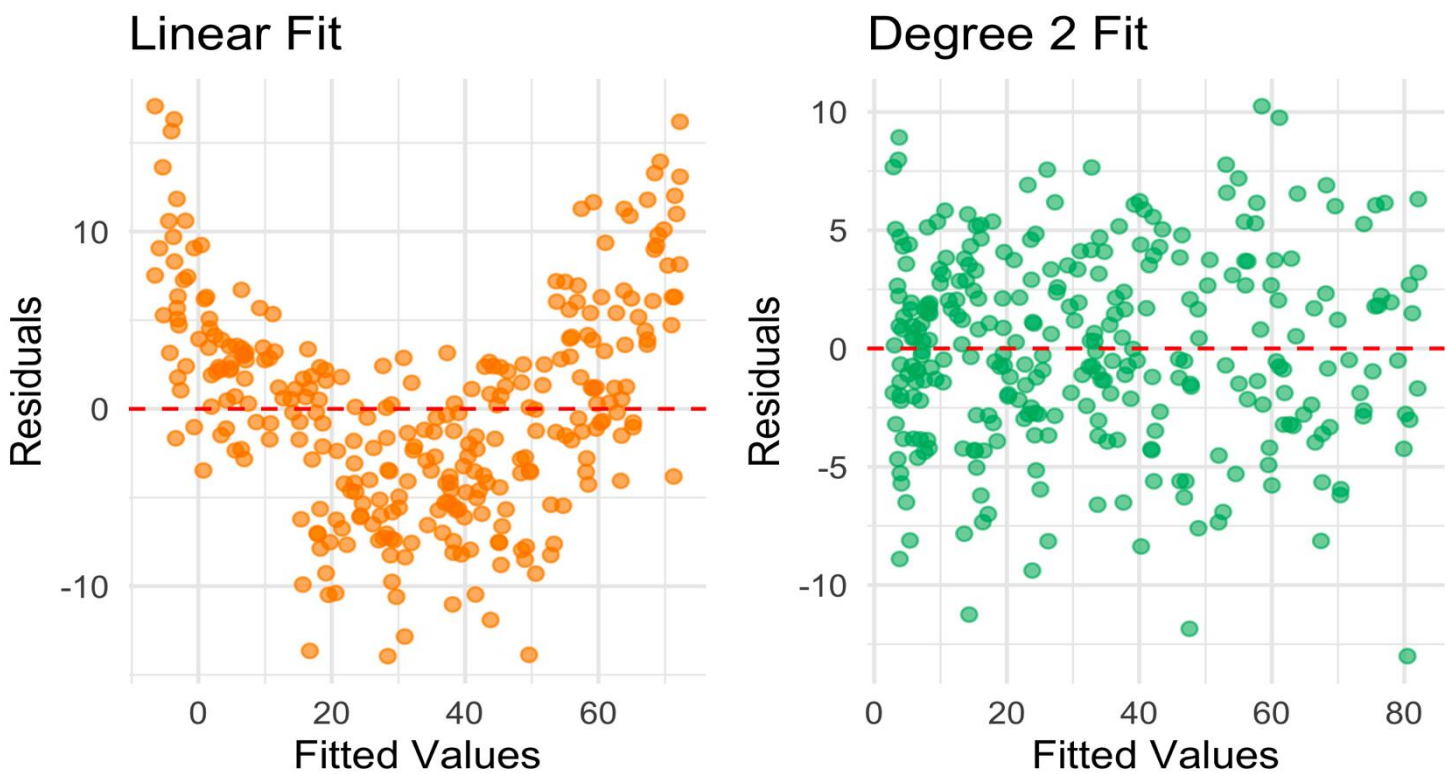
Ideally, the residual plot will show no discernible pattern. The presence of a pattern may indicate a problem with some aspect of the linear model.





Simple solution

If the residual plot indicates that there are non-linear associations in the data, then a simple approach is to use non-linear transformations of the predictors, such as $\log(X), \sqrt{X}$, and X^2, in the regression model.



Polynomial Regression

The standard way to extend linear regression to accommodate non-linear a relationship between the predictor and response has been to replace the standard linear model:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$

with a **polynomial function**

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \beta_3 X_i^3 + \cdots + \beta_d X_i^d + \epsilon_i$$

Generally it is unusual to use d greater than 3 or 4. For large degrees d, the polynomial curve can become overly flexible and start to model noise.

Polynomial Regression

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \beta_3 X_i^3 + \cdots + \beta_d X_i^d + \epsilon_i$$

- The coefficients here can be easily estimated using least squares because this is still considered a standard linear model.
- Importantly, this means that all the inference tools for linear models (standard errors, F-tests, etc.) are all available in this setting.

Examples

See R script MLR_OtherConsidertions_Part2.R